C25 Optical Isomerism – Knowledge Organiser

AQA A-level Chemistry • Set 1: Optical Isomerism **Paper 2**

#	Question	Answer
1	Chiral centre	Carbon with four different groups
2	Enantiomers	Non-superimposable mirror images
3	How enantiomers differ	Rotate plane-polarised light in opposite directions
4	Racemic mixture	Equal amounts of each enantiomer
5	Optical activity of a racemic mixture	None (rotations cancel)
6	Planar intermediates that give racemates	Carbocation, C=O group
7	Why a planar intermediate gives a racemate	Equal chance of attack from either side
8	Symbol used to mark a chiral centre	
9	Problem with racemic drugs	One enantiomer may be inactive or harmful
10	Two types of stereoisomerism	E–Z and optical